Polylogarithmic matching at spatial infinity: The massless scalar field

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Motivations

[1]

We study the asymptotic structure of a massless scalar field in flat spacetime, asking what happens when the standard fall-off conditions — almost universally assumed in the literature — are relaxed.

Logarithmic terms have surfaced repeatedly in recent work on asymptotic behaviour, suggesting the standard fall-offs may be too restrictive. Such terms appear in:

- subleading soft theorems [Campiglia,Laddha (2016) / Cachazo,Strominger (2014)]
- the Q_l branch of massless-field solutions [Fuentealba, Henneaux (2025) / Henneaux, Troessaert (2018)]
- logarithmic u-terms at null infinity [Briceño, González, Henneaux, Pérez (2026)]
- gravitational tail-effect log corrections [Laddha, Sen (2018) / Galley, Leibovich, Porto, Ross (2016)]

Set up

[3]

We consider a massless scalar field in flat four dimensional Minkowski spacetime, where the Hamiltonian action is given by

$$S[\phi,\pi]=\int dt\int d^3x\left(\pi_\phi\dot{\phi}-\frac{1}{2}\pi^2-\frac{1}{2}\eta^{ij}\partial_i\phi\partial_j\phi\right)$$

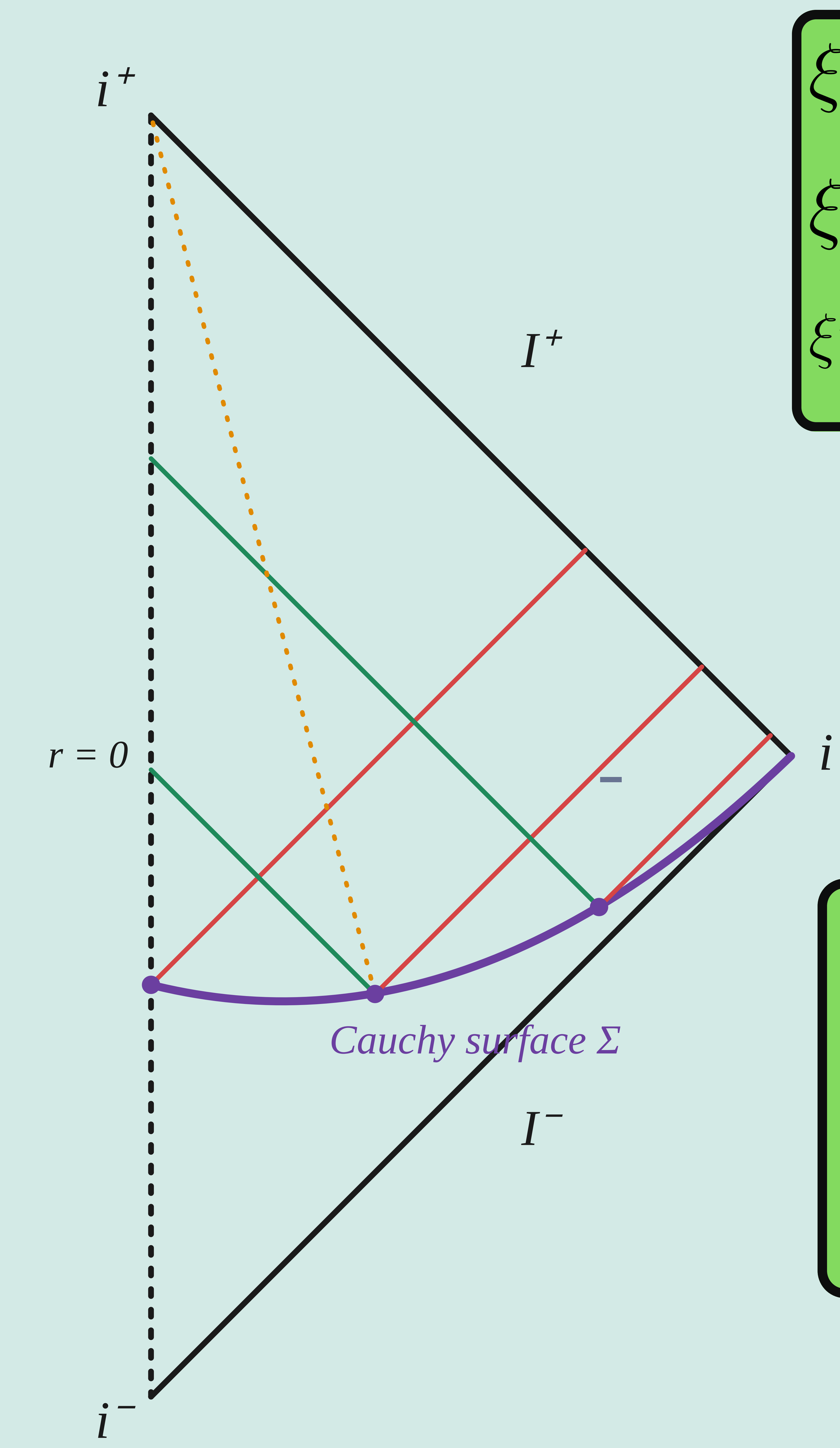
The set of asymptotic conditions is relaxed with polylogarithmic terms on the fields

$$\phi(t,r,\bar{x}^A)=\sum_{n=0}^{N}\frac{\log^n r}{r}\phi^{(n)}_{log}(t,\bar{x}^A)+\frac{\bar{\phi}}{r}(t,\bar{x}^A)+o(r^{-2})\qquad \pi(t,r,\bar{x}^A)=\sum_{m=0}^{N}\log^m r\,\pi^{(m)}_{log}(t,\bar{x}^A)+\bar{\pi}(t,\bar{x}^A)+\frac{\pi^{(1)}(t,\bar{x}^A)}{r}+o(r^{-1})$$

Can this hipotesis reveal new asymptotic structures? cambiar serie
The scalar field is the principal playground to reveal that!

The causal structure

[2]



Initial data on Σ: outgoing null rays → I^+; ingoing null rays → $r = 0$; timelike geodesic → i^+; spacelike slice Σ → i^0

$$\xi^t=br+T$$
$$\xi^r=W$$
$$\xi^A=Y^A+\frac{1}{r}D^AW$$

$$\bar{D}_A\bar{D}_BW+\bar{g}_{AB}W=0$$
$$\bar{D}_A\bar{D}_Bb+\bar{g}_{AB}b=0$$
$$\bar{D}_A\bar{Y}_B+\bar{D}_B\bar{Y}_A=0$$

Methodology and First Results

[4]

Symplectic structure

In order to stablish whether this relaxed fall-off is admissible to enlarge the asymptotic structure, the symplectyc form must fullfill the following conditions :

- Finiteness of the symplectic form Ω
- Poincaré invariance of the symplectic form $\mathcal{L}_\xi\Omega=0$
- Poincaré invariant fall-off
- Finite canonical generators $i_\xi\Omega=d_VQ_\xi$

Both set of fields $\phi^{(n)}$ and $\pi^{(n)}$ satisfy opposite strict parity conditions under the antipodal map of the sphere $x^A\to-\bar{x}^A$

Smeared charges :

$$Q^{(n)}_{\epsilon,\eta}=\oint d^2x\left(\sqrt{g}\,\epsilon^{(n)}\phi^{(n)}+\eta^{(n)}\pi^{(n)}\right)$$

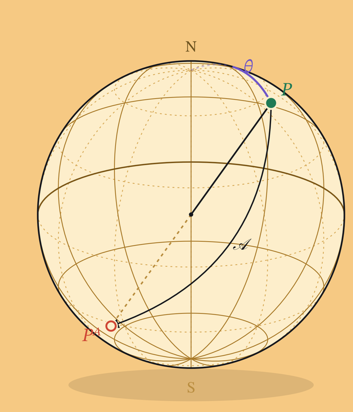


FIG. 1 · THE CELESTIAL SPHERE S²

Antipodal matching at null infinity

A direction on the celestial sphere at past null infinity I^- is identified with its *antipode* at future null infinity I^+.

- P on I^- (past null infinity)
- P^A on I^+ (future null infinity)

$$(\theta,\varphi)\mapsto(\pi-\theta,\varphi+\pi)$$

THE ANTIPODAL MAP $\mathscr{A}$ ON THE CELESTIAL SPHERE

In the analysis of asymptotic symmetries (BMS), radiative data at I^- is matched to data at I^+ through the antipodal map. This identification renders the associated charges conserved and underlies the equivalence between soft theorems and asymptotic symmetries.

Solutions at i^0

The **hyperbolic foliation** makes direct contact with the Cauchy surface while resolving the critical sets at spatial infinity

$$ds^2=d\eta^2+\eta^2h_{ab}dx^adx^b$$

$$h_{ab}=-\frac{1}{(1-s^2)^2}ds^2+\frac{1}{(1-s^2)}\bar{g}_{AB}d\bar{x}^Ad\bar{x}^B$$

On the Cauchy surface, the two descriptions of the field match:

$$\varphi^{(0)}\Big|_{s=0}=\phi^{(0)}(\bar{x}^A)\quad,\quad\varphi^{(0)}_{\log}\Big|_{s=0}=\phi^{(1)}(\bar{x}^A)$$

Translating the fall-off into a hyperbolic expansion:

$$\phi(\eta,x^a)=\sum_{k=0}^{\infty}\frac{1}{\eta^{1+k}}\left[\varphi^{(k)}(x^a)+\log\eta\,\varphi^{(k)}_{\log}(x^a)\right]$$

Prove that if x≡y mod n, then kx≡ky mod n for all integers k. Please show all work neatly

The equation for the log's ovedecouples; its solution splits into two branches:

$$(1-s^2)\partial_s^2\varphi^{(k)}_{\log}-\bar{D}^A\bar{D}_A\varphi^{(k)}_{\log}-\frac{(k^2-1)}{1-s^2}\varphi^{(k)}_{\log}=0$$

Where the general solution is known and given by

$$\varphi^{(k)}_{\log}(s,\bar{x}^A)=\varphi^{P(k)}_{\log}(s,\bar{x}^A)+\varphi^{Q(k)}_{\log}(s,\bar{x}^A)$$

Here each part is given by

$$\varphi^{P(k)}_{\log}(s,\bar{x}^A)=(1-s^2)^{\frac{1-k}{2}}\sum_{l,m}\Phi^{P(k)}_{lm}\tilde{P}^{k+\frac{1}{2}}_{l-k}(s)Y_{lm}(\bar{x}^A)$$

$$\varphi^{Q(k)}_{\log}(s,\bar{x}^A)=(1-s^2)^{\frac{1-k}{2}}\sum_{l,m}\Phi^{Q(k)}_{lm}\tilde{Q}^{k+\frac{1}{2}}_{l-k}(s)Y_{lm}(\bar{x}^A)\,,$$

At each order the log field acts as a source for the polynomial field $\varphi^{(k)}$

$$(1-s^2)\partial_s^2\varphi^{(k)}-\bar{D}^A\bar{D}_A\varphi^{(k)}-\frac{(k^2-1)}{1-s^2}\varphi^{(k)}=-\frac{2k}{1-s^2}\varphi^{(k)}_{\log}$$

The solution is also has two branches

$$\rho^{(k)}_{lm}(s)=\rho^{P(k)}_{lm}(s)+\rho^{Q(k)}_{lm}(s)$$

The solution is also has two branches, and their explicit form and recursion relations depends on the values of some constants

$$\rho^{Q(k)}_{lm}=-\log\left(\frac{1+s}{1-s}\right)\frac{\Phi^{Q(k)}_{lm}}{2a_{(k+\frac{1}{2})}}\tilde{P}^{(k+\frac{1}{2})}_{l-k}(s)+\sum_{j=0}^{3k-l-3}g^{Q(k)}_{[j],lm}s^j$$

$$\rho^{P(k)}_{lm}=-\log\left(\frac{1+s}{1-s}\right)\frac{a_{(k+\frac{1}{2})}\Phi^{P(k)}_{lm}}{2}\tilde{Q}^{(k+\frac{1}{2})}_{l-k}(s)+\sum_{j=0}^{3k-l-4}g^{P(k)}_{[j],lm}s^j$$

Work in Progress

There are a few things to check still in the scalar field

- Finish the last remaining solution case
- Match the i^0data to null infinity $\mathscr{I}^{\pm}$— compute the matching conditions
- Connect with log soft theorems

Can be done in other theories?

- Maxwell (spin-1) field: preliminary results are affirmative
- Kalb–Ramond two-form: the 4D dual of the scalar
- Gravity: less clear — still an open question

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